FRC 9470 2025 REEFSCAPE

SUMMARY

REEFSCAPE transported teams into an underwater ecosystem, where alliances collected and placed “coral” (4 inch PVC pipe) into reef structures and delivered “algae” (16 inch bouncy balls) to designated zones on a “reef”. The goal for both of our robots this year was to score on all levels of the reef with a stationary end effector as well as being able to take algae off the reef to allow other teams to score.

Robot 1 for Pinnacles Regionals 2025

- Used WCP Swerve X2Ts to optimize our robot’s design; these module’s profiles allowed us to have more space inside the robot for electronics and mechanisms. Featured a belt-driven funnel intake, acting like a conveyor system to guide coral into a horizontal roller end effector. The end effector was machined out of aluminum and mounted on a two-stage cascade elevator. This allowed for us to raise our end effector quickly in order to score all reef layers very fast.
- Ranked 15th overall and was first pick of Alliance 3, advancing to quarterfinals and winning the Autonomous award
- For this robot, I led the design process, creating the layout sketches and mainly working on the end effector, elevator and drivetrain. I also helped schedule the entire build process and manage the order list spreadsheet.
- This event was a huge milestone for our team as it was our best overall performance at the time and was a massive motivator for our rebuild and future competitions.
- Testing Video



Robot 2 for East Bay Regional 2025

- Used the same WCP Swerve X2Ts to optimize our robot's design; these module's profiles allowed us to have more space inside the robot for electronics and mechanisms. The end effector was significantly smaller and used bottom rollers and only one motor, reducing weight to a little over two pounds and drastically improving elevator speed. (compared to our almost 7 pound end effector at pinnacles). Introduced a gravity-powered funnel with 3D-printed star wheels for automatic coral alignment and smooth feeding into the end effector. This design offered a larger intake area and steeper angle, allowing faster, more reliable gameplay compared to the powered funnel at Pinnacles.
- Ranked 16th and joined Alliance 6, reaching round 3 of playoffs.
- Overall, this robot was a lighter, faster, and more optimized version of its predecessor, our Pinnacles robot and this time we were not 4 pounds overweight on load in day!
- Despite our worse performance, this robot still stays as my favorite robot built on 9470





